

Računarske vežbe - Sistemi automatskog upravljanja

Veza položaja polova i ponašanja sistema

Podsećanje

- Sistemi prvog reda - aperiodičan odziv
- Sistemi višeg reda:
 - realni polovi - aperiodičan odziv
 - konjugovano-kompleksni polovi - oscilatoran odziv
- Realan deo pola utiče na stabilnost sistema i brzinu smirenja odziva
- Imaginaran deo pola utiče na učestanost oscilovanja

1. Odrediti dominantne polove sledećih sistema:

a) $G(s) = \frac{s+3}{(s+2)(s+1)}$

$p = -1$

b) $G(s) = \frac{1}{(s+3)(s+(2-j))(s+(2+j))}$

$p = -2 \pm j$

c) $G(s) = \frac{s+1}{(s+5)(s+6)}$

$p = -5$

2. Koji sistem brže ulazi u ustaljeno stanje?

a) $G_1(s) = \frac{2}{s+1}$

ili $G_2(s) = \frac{1}{s+3}$

b) $G_1(s) = \frac{1}{s+2}$

ili $G_2(s) = \frac{3}{s+4}$

c) $G_1(s) = \frac{s+2}{s^2+7s+12}$

ili $G_2(s) = \frac{5}{s^2+5s+6}$

d) $G_1(s) = \frac{1}{(s-1)(s-6)}$

ili $G_2(s) = \frac{1}{(s-5)(s-7)}$

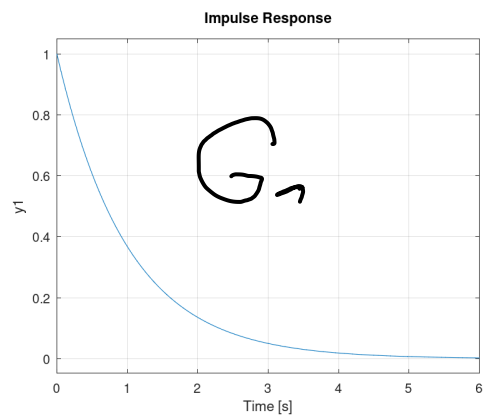
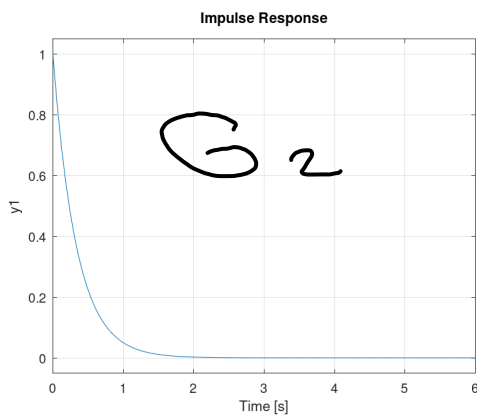
e) $G_1(s) = \frac{1}{s^2+6s+9}$

ili $G_2(s) = \frac{1}{s^2+2s+5}$

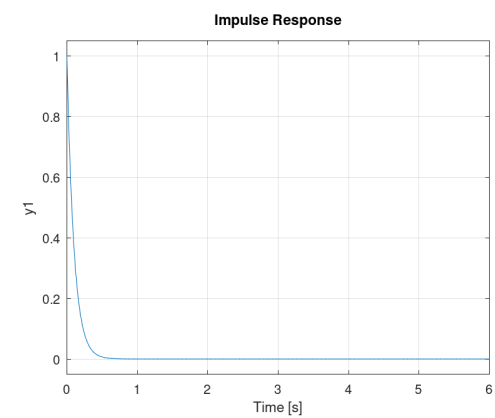
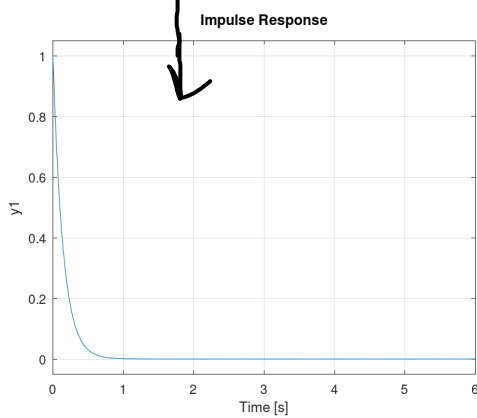
nestabilni
 $\Rightarrow$ ne ulaze u
ustaljeno
stanje

3. Koji impulsni odziv odgovara kom sistemu/odzivu/polu?

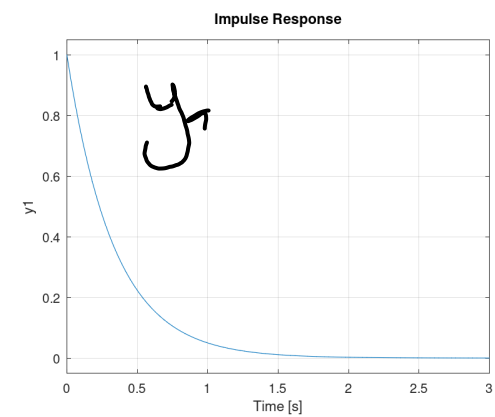
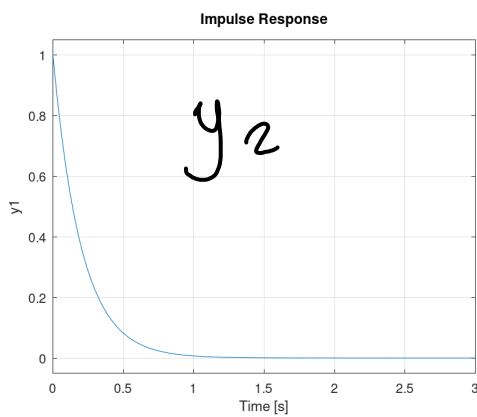
a) $G_1(s) = \frac{1}{s+1}$ $G_2(s) = \frac{1}{s+3}$



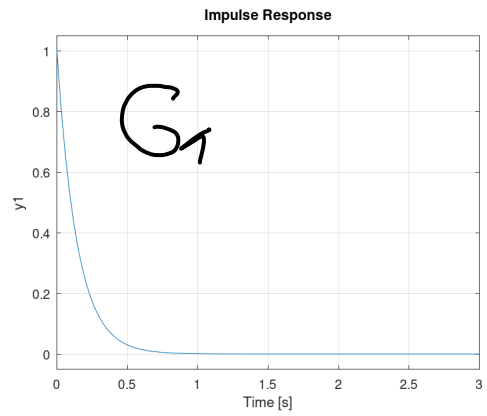
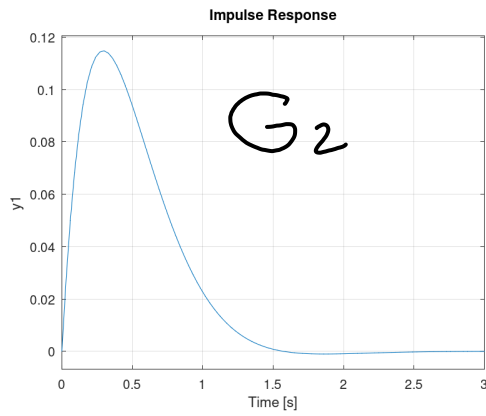
b) $G_1(s) = \frac{1}{s+7}$ $G_1(s) = \frac{1}{s+10}$



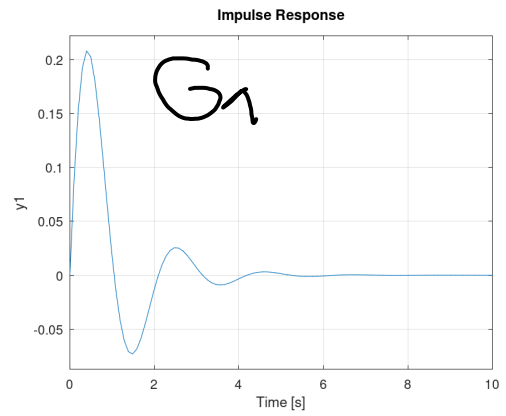
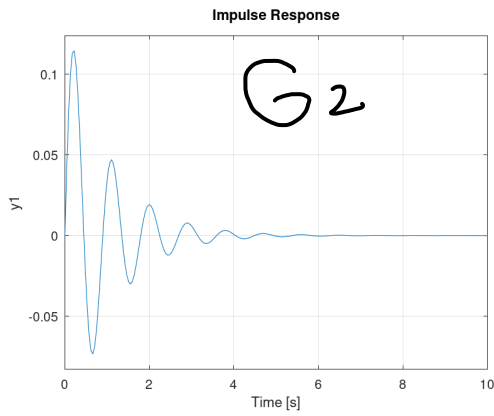
c) $y_1(t) = e^{-3t}h(t)$ $y_2(t) = e^{-5t}h(t)$



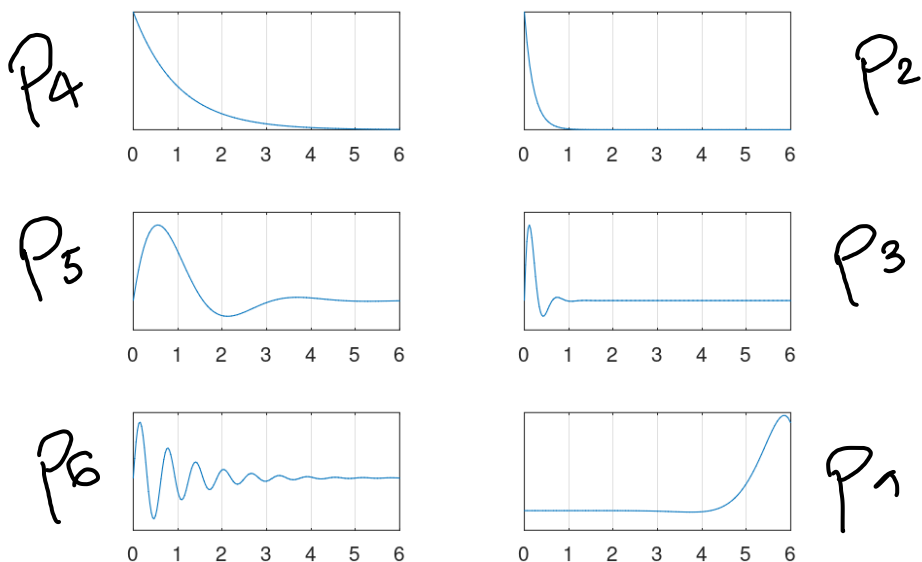
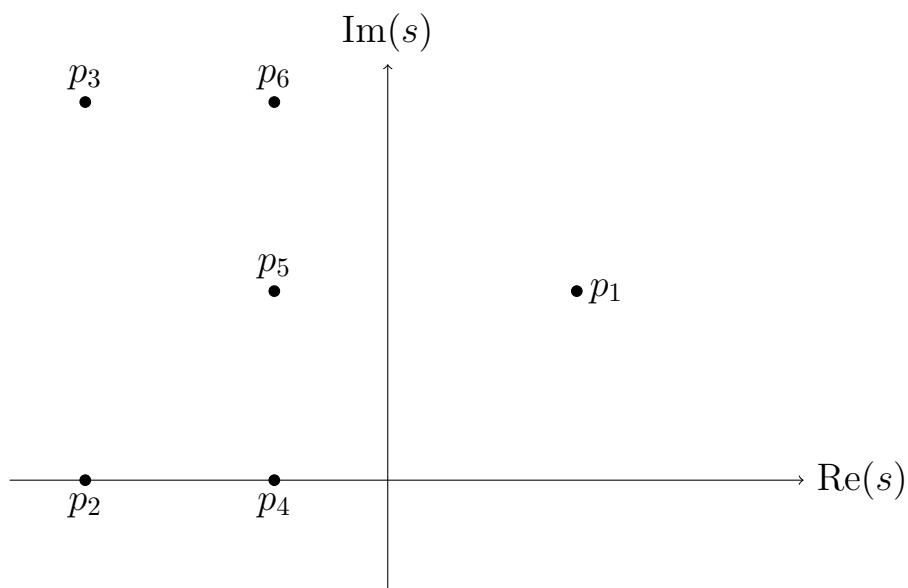
d) $G_1(s) = \frac{1}{s+7}$ $G_2(s) = \frac{1}{(s+(3-2j))(s+(3+2j))}$



e) $G_1(s) = \frac{1}{(s+(1+3j))(s+(1-3j))}$ $G_2(s) = \frac{1}{(s+(1+7j))(s+(1-7j))}$



f) Dati su položaji polova



4. Izračunati dominantnu vremensku konstantu sledećih sistema:

a) $G(s) = \frac{1}{s+3}$

$$T_d = \frac{1}{|-3|} = \frac{1}{3} \text{ s}$$

b) $G(s) = \frac{s+3}{(s+2)(s+5)}$

$$T_d = \frac{1}{2} \text{ s}$$

c) $G(s) = \frac{2}{(s+(4+j))(s+(4-j))}$

$$T_d = \frac{1}{4} \text{ s}$$

d) $G(s) = \frac{1}{(s-3)(s-4)}$

nestabilan $\Rightarrow$ ne ulazi u ustaljeno stanje

5. Izračunati statičko pojačanje sledećih sistema:

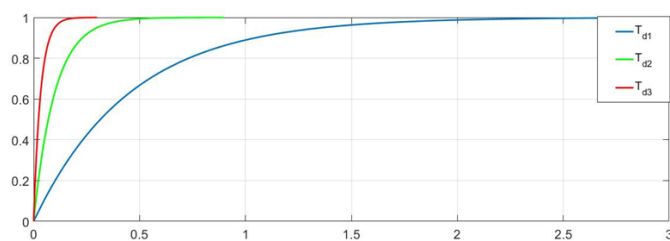
a) $G(s) = \frac{1}{s^2+3s+9}$

$$K = \lim_{s \rightarrow 0} G(s) = \frac{1}{9}$$

b) $G(s) = \frac{3}{s^3+s^2+3s+7}$

$$K = \lim_{s \rightarrow 0} G(s) = \frac{3}{7}$$

6. Poređati dominantne vremenske konstante sledećih odziva od najmanje do najveće



$$T_{d3} < T_{d2} < T_{d1}$$